

CSA3006 - DATA MINING AND DATA WAREHOUSING

Presented By

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Module I

- **Data Warehouse:** Basic Concepts, Differences between Operational Database Systems and Data Warehouses- A Multitiered Architecture - Data Warehouse Models: Extraction, Transformation and Loading - Metadata Repository -Data Cube and OLAP -Data Warehouse Design and Usage – Data warehouse implementation.
- **Introduction To Data Mining:** Introduction in knowledge discovery from data. Basic concepts and main data mining tasks. Steps in knowledge discovery from database process -Architecture of a data mining systems - Data mining on different kinds of data - Different kinds of pattern - Technologies used - Applications – KEEL (Knowledge Extraction based on Evolutionary Learning) tool. Data categories and types of attributes.

Module II

- **Data Preprocessing:** Data Objects and attribute types - Basic statistical description of data - Data visualization – Measuring data similarity and dissimilarity - Data cleaning - Integration - Data reduction – Data transformation and data discretization.
- **Association Rule Mining:** Basic concepts - Frequent itemset mining methods - Apriori algorithm, Pattern growth approach for mining frequent itemsets, Mining frequent itemsets using vertical data format, Mining closed and max patterns - Pattern mining in multilevel and multidimensional space.

Module III

- **Classification:** Basic concepts—Decision tree induction—Bayesian classification, Rule-based classification, Support Vector Machines, Improve Classification Accuracy, Model Evaluation and Selection - Techniques to Improve Classification Accuracy.
- **Regression Analysis:** Linear and Logistic Regression, Neural Networks, Regression Evaluation.

Module IV

Clustering and Applications: Basic concepts (cluster, centroid), Cluster analysis—Types of Data in Cluster Analysis—Categorization of Major Clustering Methods— algorithms (KMeans, Fuzzy CMeans), Hierarchical Methods— Density-Based Methods, Statistical-based clustering (EM algorithm). Graph-Based Methods, Spectral techniques Outlier Analysis.

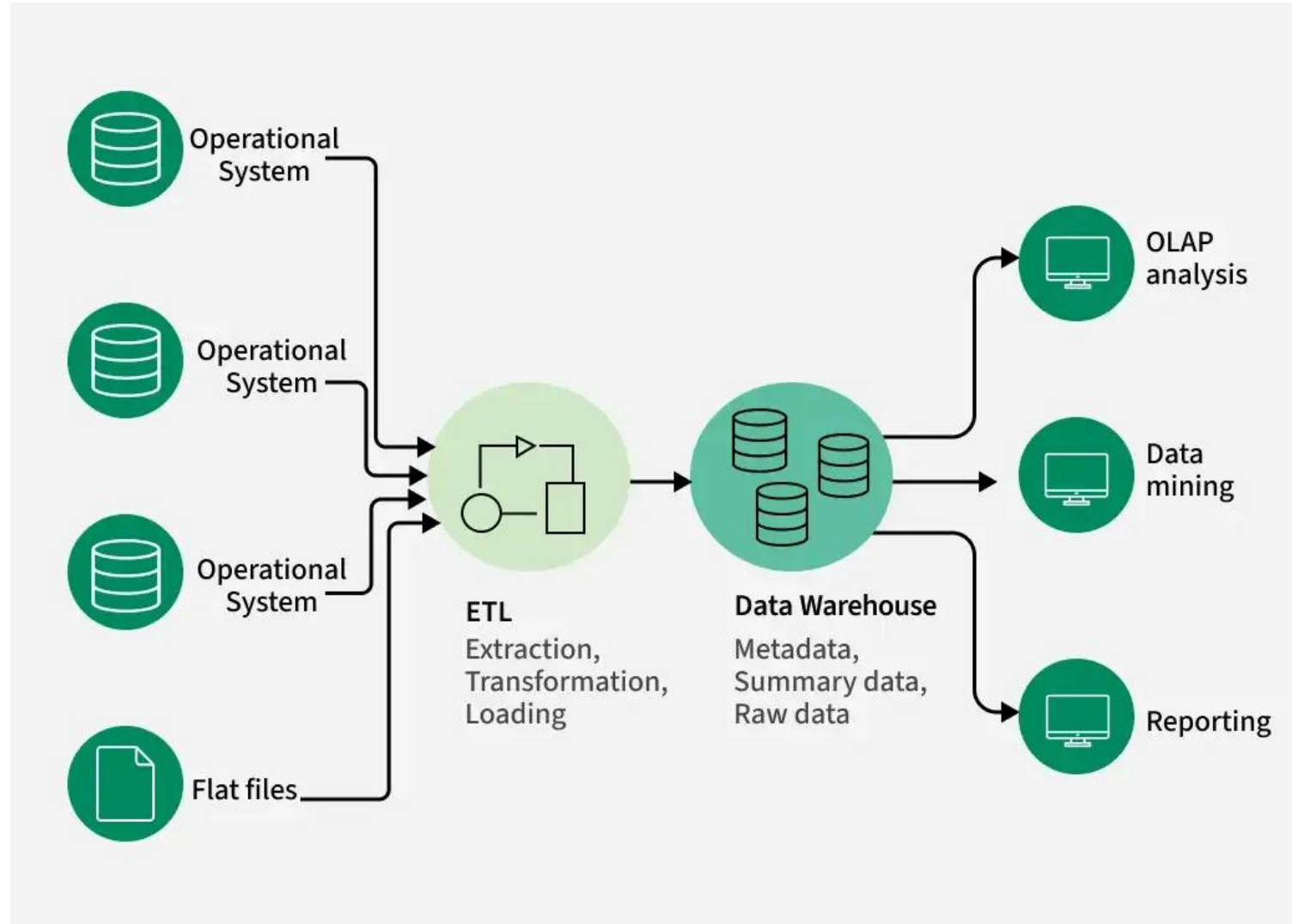
Module V

- **Advanced Data Mining-** Regression and time series processing. Nonlinear regression models. Regression trees. Radial Basis Networks. Time series analysis (trend analysis, pattern detection, prediction models, anomaly detection). Ensemble methods. Voting. Bagging. Boosting (AdaBoost). Random forests. Stacking. Mining sequence patterns in Transactional databases– Mining Object– Spatial– Multimedia–Text and Web data – Spatial Data mining– Multimedia Data mining–Text Mining– Mining the World Wide Web. Document classification and clustering.
- **Case Study:** Mining Real data- Pre-processing data from a real medical domain. Applying various data mining techniques to create a comprehensive and accurate model of the data.

Data warehousing – Basic Concept

- Data warehousing is the process of *collecting, integrating, storing and managing data* from multiple sources in a central repository. It enables organizations *to organize large volumes of current and historical data* for *efficient querying, analysis and reporting*.
- The main goal of data warehousing is *to support decision-making by providing clean, consistent and timely access to data*. It ensures fast data retrieval even when working with massive datasets.

Data Warehouse Architecture



Need for Data Warehousing

- **Handling Large Data Volumes:** Traditional databases store limited data (MBs to GBs), while data warehouses are built to handle huge datasets (up to TBs), making it easier to store and analyze long-term historical data.
- **Enhanced Analytics:** Databases handle transactions; data warehouses are optimized for complex analysis and historical insights.
- **Centralized Data Storage:** A data warehouse combines data from multiple sources, giving a single, unified view for better decision-making.
- **Trend Analysis:** By storing historical data, a data warehouse allows businesses to analyze trends over time, enabling them to make strategic decisions based on past performance and predict future outcomes.
- **Business Intelligence Support:** Data warehouses work with BI tools to give quick access to insights, helping in data-driven decisions and improving efficiency.

Components of Data Warehouse

- **Data Sources:** These are the various operational systems, databases and external data feeds that provide raw data to be stored in the warehouse.
- **ETL (Extract, Transform, Load) Process:** The ETL process is responsible for extracting data from different sources, transforming it into a suitable format and loading it into the data warehouse.
- **Data Warehouse Database:** This is the central repository where cleaned and transformed data is stored. It is typically organized in a multidimensional format for efficient querying and reporting.
- **Metadata:** Metadata describes the structure, source and usage of data within the warehouse, making it easier for users and systems to understand and work with the data.
- **Data Marts:** These are smaller, more focused data repositories derived from the data warehouse, designed to meet the needs of specific business departments or functions.
- **OLAP (Online Analytical Processing) Tools:** OLAP tools allow users to analyze data in multiple dimensions, providing deeper insights and supporting complex analytical queries.
- **End-User Access Tools:** These are reporting and analysis tools, such as dashboards or Business Intelligence (BI) tools, that enable business users to query the data warehouse and generate reports.

Types of Data Warehouses

- The different types of Data Warehouses are:
- **Enterprise Data Warehouse (EDW):** A centralized warehouse that stores data from across the organization for analysis and reporting.
- **Operational Data Store (ODS):** Stores real-time operational data used for day-to-day operations, not for deep analytics.
- **Data Mart:** A subset of a data warehouse, focusing on a specific business area or department.
- **Cloud Data Warehouse:** A data warehouse hosted in the cloud, offering scalability and flexibility.
- **Big Data Warehouse:** Designed to store vast amounts of unstructured and structured data for big data analysis.
- **Virtual Data Warehouse:** Provides access to data from multiple sources without physically storing it.
- **Hybrid Data Warehouse:** Combines on-premises and cloud-based storage to offer flexibility.
- **Real-time Data Warehouse:** Designed to handle real-time data streaming and analysis for immediate insights.

Data Warehouse vs DBMS

Database	Data Warehouse
A common database is based on operational or transactional processing. Each operation is an indivisible transaction.	A data warehouse is based on analytical processing.
Generally, a database stores current and up-to-date data which is used for daily operations.	A data warehouse maintains historical data over time. Historical data is kept over years and can be used for trend analysis, future predictions, and decision support.
A database is generally application-specific.	A data warehouse is generally integrated at the organizational level by combining data from different databases.
Example: A database stores related data, such as student details in a school.	Example: A data warehouse integrates data from one or more databases so that analysis can be done to obtain results, such as identifying the best-performing school in a city.
Constructing a database is not very expensive.	Constructing a data warehouse can be expensive.

Issues Occur while Building the Warehouse

1. When and How to Gather Data?

- Source-driven: Data sources push updates to the warehouse periodically or continuously.
- Destination-driven: The warehouse pulls data on a fixed schedule.
- Perfect sync is costly, so data is slightly outdated - acceptable for analysis.

2. What Schema to Use?

- Sources have varied formats.
- The warehouse stores a cleaned, unified version - not a direct copy, but a consistent snapshot for analysis.

3. Data Transformation and Cleansing

- Fixes errors like typos or invalid codes using reference data.
- Fuzzy lookup helps match similar but not identical values.

4. How to Propagate Updates?

- If warehouse schema = source schema -> easy sync.
- If not -> it becomes a view maintenance challenge.

5. What Data to Summarize?

- Raw data is large; store summaries (e.g., total sales by category).
- Aggregates support efficient querying without full details.

Real world Example of Data warehousing

1. E-commerce: Flipkart

- **Data Gathering:** Orders, returns, payments, user clicks, delivery updates.
- **Schema:** Combines source data into a structured star schema for analysis.
- **Cleansing:** Standardizes customer names, locations and product categories.
- **Updates:** Near real-time or scheduled loads for fresh insights.
- **Summarization:** Bestsellers by category, regional demand trends, logistics performance.

2. Banking: HDFC Bank

- **Data Gathering:** ATM transactions, online banking, credit card usage, loan records.
- **Schema:** Integrates data from core banking, CRM and fraud detection systems.
- **Cleansing:** Fixes inconsistencies in account info, transaction logs and addresses.
- **Updates:** Transaction data is batched and uploaded nightly.
- **Summarization:** Daily cash flow reports, high-risk account flags and customer profitability analysis.

Advantages and Disadvantages of Data Warehousing

Advantages	Disadvantages
Better Decisions: Centralized data supports faster, smarter decisions.	High Cost: Setup requires major investment.
Business Intelligence: Enables strong operational insights.	Complexity: Needs skilled professionals to manage.
High Data Quality: Ensures consistency and reliability.	Time-Consuming: Long setup and integration time.
Scalable: Handles large and growing datasets.	Integration Issues: Combining data from multiple sources can be challenging.

Differences between Operational Database Systems and Data Warehouse

Operational Database (OLTP)

- Operational Databases are the **primary source of data** for information systems
- Used to support **day-to-day business operations**
- Store **detailed, transaction-level data**
- Data is **frequently updated** to reflect current transactions
- Designed for **real-time processing**
- Also known as **OLTP (Online Transaction Processing) systems**
- Focus on **speed, accuracy, and consistency**

Data Warehouse (OLAP)

- Data Warehouses support **data analysis and decision-making**
- Used mainly by **analysts and decision-makers**
- Store **historical and consolidated data**
- Integrate data from **multiple operational databases**
- Optimized for **querying and reporting**
- Present data in **various formats** based on user needs
- Also known as **OLAP (Online Analytical Processing) systems**

Purpose Comparison

- **Operational Databases**

- Run daily business activities
- Handle high-volume transactions
- Focus on current data

- **Data Warehouses**

- Support strategic decisions
- Enable trend analysis and forecasting
- Focus on historical data

Examples

- **Operational Database Systems**
 - Customer Relationship Management (CRM)
 - Inventory Management Systems
 - Order Processing Systems
- **Data Warehouse Applications**
 - Business Intelligence (BI)
 - Data Analytics
 - Executive Decision Support

Differences between operational database systems and data warehouses

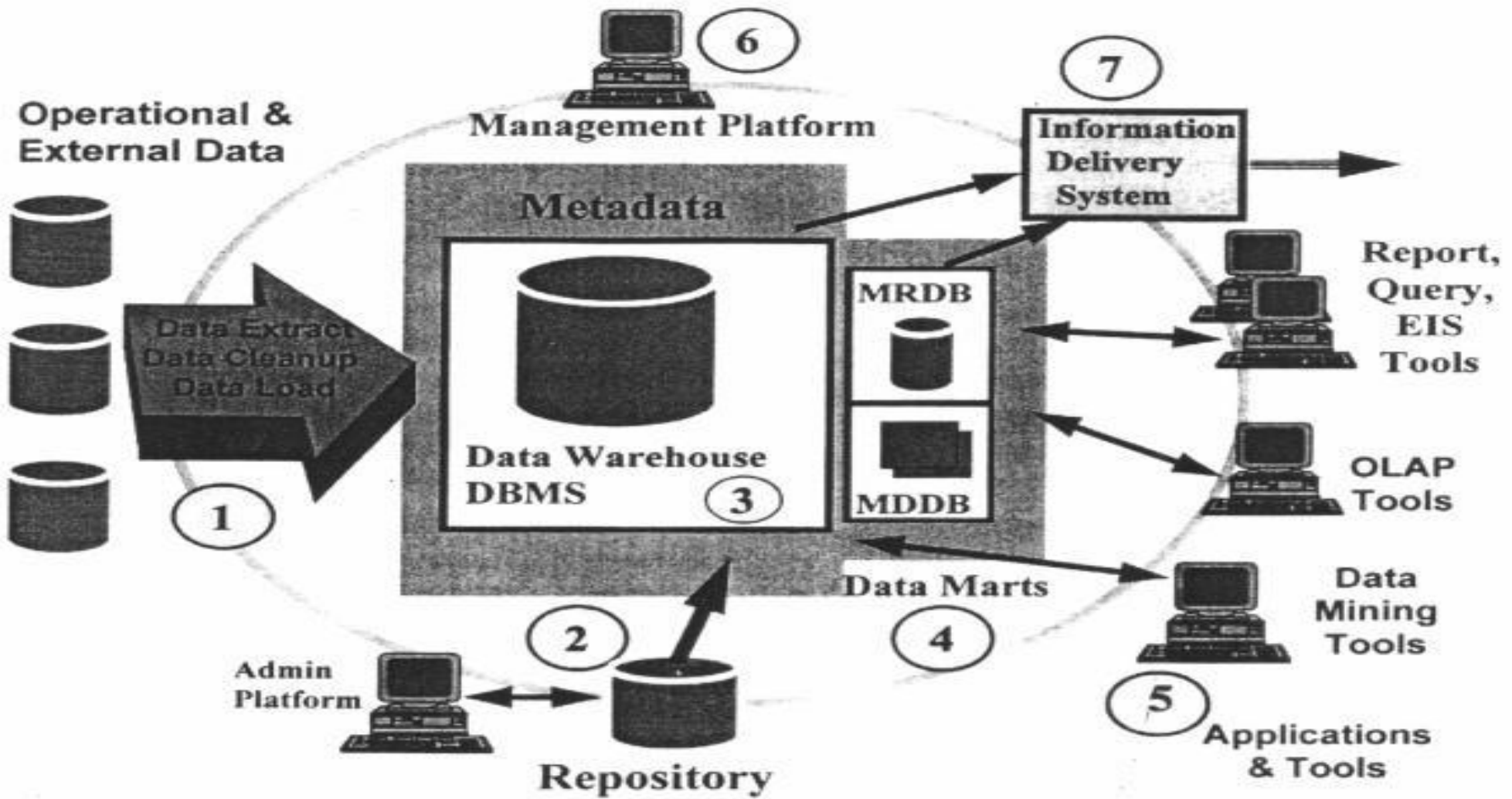
Aspect	Operational Database Systems	Data Warehouses
Purpose	Support day-to-day operations of an organization	Support decision-making and analytical activities
Data Structure	Normalized structure with multiple related tables to reduce redundancy and ensure consistency	Denormalized structure with fewer tables optimized for reporting and analysis
Data Volume	Store smaller volumes of data, mainly current and recent data	Store large volumes of data, including years of historical data
Performance	Optimized for high-speed, high-volume transaction processing	Optimized for complex queries, reporting, and analytical processing

Operational Database	Data Warehouse
Operational frameworks are designed to support high-volume transaction processing.	Data warehousing frameworks are designed to support high-volume analytical processing (OLAP).
Operational systems primarily deal with current data.	Data warehousing systems mainly handle historical and verifiable data.
Data in operational systems is frequently updated to meet real-time needs.	Data is non-volatile and changes infrequently once loaded.
Designed for real-time business operations and process management.	Designed for analysis of business measures by subject areas, categories, and dimensions.
Relational databases are designed for Online Transaction Processing (OLTP).	Data warehouses are designed for Online Analytical Processing (OLAP).
Optimized for fast inserts and updates of relatively small data volumes.	Optimized for fast retrieval and analysis of large data volumes.
Data is mainly input-oriented.	Data is mainly output-oriented.
Operational databases are generally application-oriented.	Data warehouses are generally subject-oriented.

A Multitiered Architecture

1. Overall Architecture

- The data warehouse architecture is based on a **relational database management system server** that functions as the central repository for informational data. The data warehousing architecture shown in fig 1 clearly identifies seven data warehouse components:
 1. Data sourcing, cleanup, transformation, and migration tools
 2. Metadata repository
 3. Warehouse/database technology
 4. Data marts
 5. Data query, reporting, analysis, and mining tools
 6. Data warehouse administration and management
 7. Information delivery system



Data warehouse environment.

Fig 1: Data Warehouse Architecture

- Data warehouse is an environment, not a product which is based on relational database management system that functions as the central repository for informational data.
- The central repository information is surrounded by number of key components designed to make the environment is functional, manageable and accessible.
- The data source for data warehouse is coming from operational applications. The data entered into the data warehouse transformed into an integrated structure and format. The transformation process involves conversion, summarization, filtering and condensation. The data warehouse must be capable of holding and managing large volumes of data as well as different structure of data structures over the time.

2. Data warehouse database

- This is the central part of the data warehousing environment. This is the item number 2 in the above architecture diagram. This is implemented based on RDBMS technology.
- The data warehouse attributes are:
 - Very large database size
 - ad hoc query processing
 - need for flexible user view creation including aggregates, multi table joins and drill-downs.
- These approaches include:
 - Parallel relational database designs for scalability that includes shared-memory, shared disk or shared nothing models implemented.
 - An innovative approach to speed up a traditional RDBMS.
 - Multidimensional database (MDDBS) that are based on proprietary database technology.

3. Sourcing, Acquisition, Clean up, and Transformation Tools

- This is item number 1 in the above arch diagram. They perform conversions, summarization, key Changes, structural changes and condensation.
- The data transformation is required so that the information can be used by decision support tools. The transformation produces programs, control statements, JCL code, COBOL code, UNIX scripts, and SQL DDL code etc., to move the data into data warehouse from multiple operational systems.
- *The functionalities of these tools are listed below:*
 - *Removing* unwanted data from operational database
 - *Converting* to common data names and attributes
 - *Calculating* summaries and derived data
 - *Establishing* defaults for missing data
 - *Accommodating* source data definition changes

- *Issues to be considered while data sourcing, cleanup, extract and transformation:*
- **Data heterogeneity:**
- It refers to DBMS different nature such as it may be in different data modules, it may have different access languages, it may have data navigation methods, operations, concurrency, integrity and recovery processes etc.,
- **Data heterogeneity:**
- It refers to the different way the data is defined and used in different modules.
- ***Some experts involved in the development of such tools:***
- Prism Solutions, Evolutionary Technology Inc., Vality, Praxis and Carleton

4. Meta data

- It is data about data. It is used for maintaining, managing and using the data warehouse. It is classified into two:

i. Technical Meta data:

- It contains information about data warehouse data used by warehouse designer, administrator to carry out development and management tasks. It includes,
 - Info about data stores
 - Transformation descriptions.
 - Warehouse Object and data structure definitions for target data
 - The rules used to perform clean up, and data enhancement
 - Data mapping operations
 - Access authorization, backup history, archive history, info delivery history, data acquisition history, data access etc.,

ii. Business Meta data:

- It contains info that gives info stored in data warehouse to users. It includes,
- Subject areas, and info object type including queries, reports, images, video, audio clips etc.
- Internet home pages
- Info related to info delivery system
- Data warehouse operational info such as ownerships, audit trails etc.,

Meta data helps the users to understand content and find the data. Meta data are stored in a separate data stores which is known as informational directory or Meta data repository which helps to integrate, maintain and view the contents of the data warehouse. The following lists the characteristics of information directory and metadata:

- It is the gateway to the data warehouse environment
- It supports easy distribution and replication of content for high performance and availability
- It should be searchable by business oriented key words
- It should act as a launch platform for end user to access data and analysis tools
- It should support the sharing of info
- It should support scheduling options for request
- IT should support and provide interface to other applications
- It should support end user monitoring of the status of the data warehouse environment

5. Access tools

- Its purpose is to provide information to business users for decision making. There are five categories:
 - Data query and reporting tools
 - Application development tools
 - Executive info system tools (EIS)
 - OLAP tools
 - Data mining tools

a. Query and reporting tools

- These tools are used to generate query and report. There are two types of reporting tools. They are:
- Production reporting tool used to generate regular operational reports
- Desktop report writer are inexpensive desktop tools designed for end users.
- ***Managed Query tools***
- It is used to generate SQL query. It uses Meta layer software in between users and databases which offers a point-and-click creation of SQL statement.
- This tool is a preferred choice of users to perform segment identification, demographic analysis, territory management and preparation of customer mailing lists etc.

b. Application development tools:

- This is a graphical data access environment which integrates OLAP tools with data warehouse and can be used to access all db systems

c. OLAP Tools:

- These are used to analyze the data in multi dimensional and complex views. To enable multidimensional properties it uses MDDDB and MRDB where MDDDB refers multi dimensional data base and MRDB refers multi relational data bases.

d. Data mining tools: These are used to discover knowledge from the data warehouse data also can be used for data visualization and data correction purposes.

6. Data marts

- Data mart is a data store to a data warehouse of integrated data.
- Departmental subsets that focus on selected subjects. They are independent used by dedicated user group. They are used for rapid delivery of enhanced decision support functionality to end users.
- ***Data mart is used in the following situation:***
 - Extremely urgent user requirement
 - The absence of a budget for a full scale data warehouse strategy
 - The decentralization of business needs
 - The attraction of easy to use tools and mind sized project
- *Data mart presents two problems:*
 - ***Scalability:***
 - A small data mart can grow quickly in multi dimensions. So that while designing it, the organization has to pay more attention on system scalability, consistency and manageability issues.
 - *Data integration*

7. Data warehouse administration and management

- The management of data warehouse includes,
- Security and priority management
- Monitoring updates from multiple sources
- Data quality checks
- Managing and updating meta data
- Auditing and reporting data warehouse usage and status
- Purging data
- Replicating, sub setting and distributing data
- Backup and recovery
- Data warehouse storage management which includes capacity planning, hierarchical storage management and purging of aged data etc.,

8. Information delivery system

- It is used to enable the process of subscribing for data warehouse info.
- Delivery to one or more destinations according to specified scheduling algorithm

Data Warehouse Models

- A **Data Warehouse Model** defines how data is **organized, stored, and related in a data warehouse** to support reporting, analysis, and decision-making.

1. Enterprise Warehouse

- Collects information about subjects across the **entire organization**.
- Provides **corporate-wide data integration**.
- Data is gathered from **operational systems** and **external information providers**.
- Cross-functional in scope.
- Contains both **detailed data** and **summarized data**.
- Data size ranges from **a few gigabytes to terabytes or more**.
- Can be implemented on **mainframes, super servers, or parallel architecture platforms**.
- Requires **extensive business modeling**.
- Design and implementation may take **several years**.

2. Data Mart

- Contains a **subset of corporate data** for a **specific group of users**.
- Scope is limited to **selected subjects**.

Example: A marketing data mart focuses on **customer, item, and sales**.

- Data is usually **summarized**.
- Implemented on **low-cost departmental servers**.
- Supports **UNIX/LINUX or Windows-based systems**.
- Implementation time is **short (weeks rather than years)**.
- Poor planning may cause **complex integration issues** later.

Types of Data Marts

- **Independent Data Mart**
 - Data sourced directly from **operational systems** or **external providers**.
- **Dependent Data Mart**
 - Data sourced directly from an **enterprise data warehouse**.

3. Virtual Warehouse

- A **collection of views** over operational databases.
- Only selected **summary views are materialized** for better performance.
- **Easy and fast to build.**
- Requires **high processing capacity** on operational database servers.

Components in Data Modeling

- **Fact Tables:** Contain measurable data (facts) like sales amount, quantity sold, etc. Often includes foreign keys to dimension tables.
- **Dimension Tables:** Provide context to facts, including details like customer name, product category, or geographic region.
- **Measures and Metrics:** Represent quantitative data used in analysis, such as total revenue or average order value.
- **Hierarchies:** Show levels of data granularity, such as Year → Quarter → Month in a time dimension.
- **Keys and Relationships:** Primary and foreign keys define the connections between fact and dimension tables.

Best Practices for Data Modeling in Data Warehouses

- **Understand Business Requirements:** Ensure that the data model aligns with the business goals and provides the necessary insights for decision-making.
- **Focus on Flexibility:** Design the data model to be flexible and scalable to accommodate future data growth and changes in business needs.
- **Optimize for Query Performance:** Consider the types of queries that will be run on the data warehouse and optimize the model accordingly, whether through indexing, partitioning, or denormalization.
- **Maintain Data Integrity:** Use constraints and relationships to maintain data integrity and prevent anomalies.
- **Document the Data Model:** Maintain thorough documentation of the data model to ensure that it is easily understood and maintained by others.

Schema Design

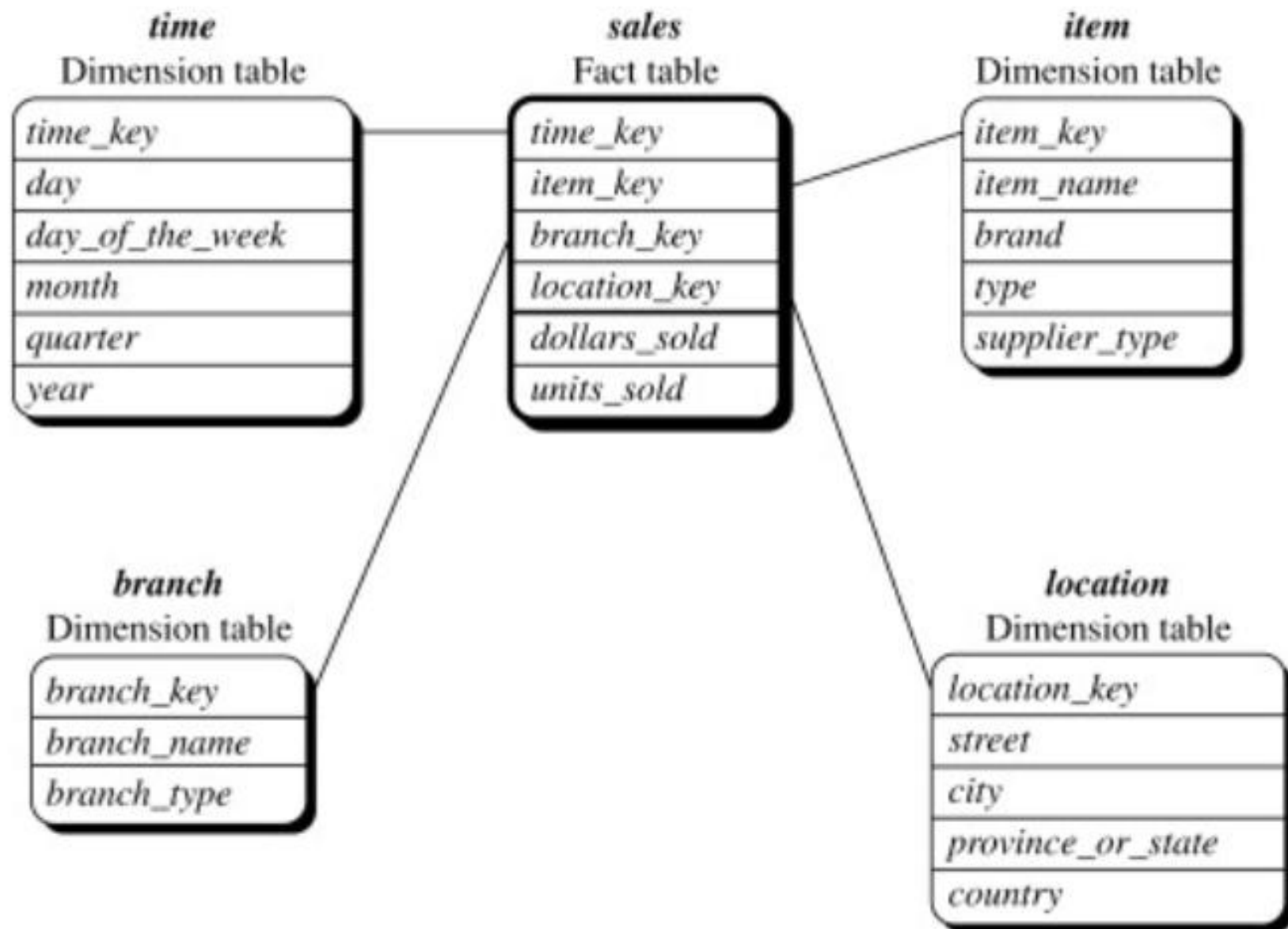
- Traditional **Entity–Relationship (ER) model** is mainly used for relational databases and supports **online transaction processing (OLTP)**.
- A **data warehouse** needs a **concise, subject-oriented schema** designed for **online analytical processing (OLAP)**.
- The most widely used approach for data warehouses is the **multidimensional data model**.
- Multidimensional schemas are classified into **Star Schema, Snowflake Schema, and Fact Constellation Schema**.

1. Star Schema

- The **most common and popular schema** used in data warehouses.
- Consists of a **central fact table** that stores quantitative data (measures).
- Fact table contains the **bulk of the data** and is designed with **no redundancy**.
- Surrounded by multiple **dimension tables**, one for each business dimension (e.g., time, product, customer).
- Dimension tables are **smaller** and store descriptive attributes.
- The structure forms a **star-like pattern**, with dimension tables arranged radially around the fact table.
- Simple design that enables **fast query performance and easy understanding**.

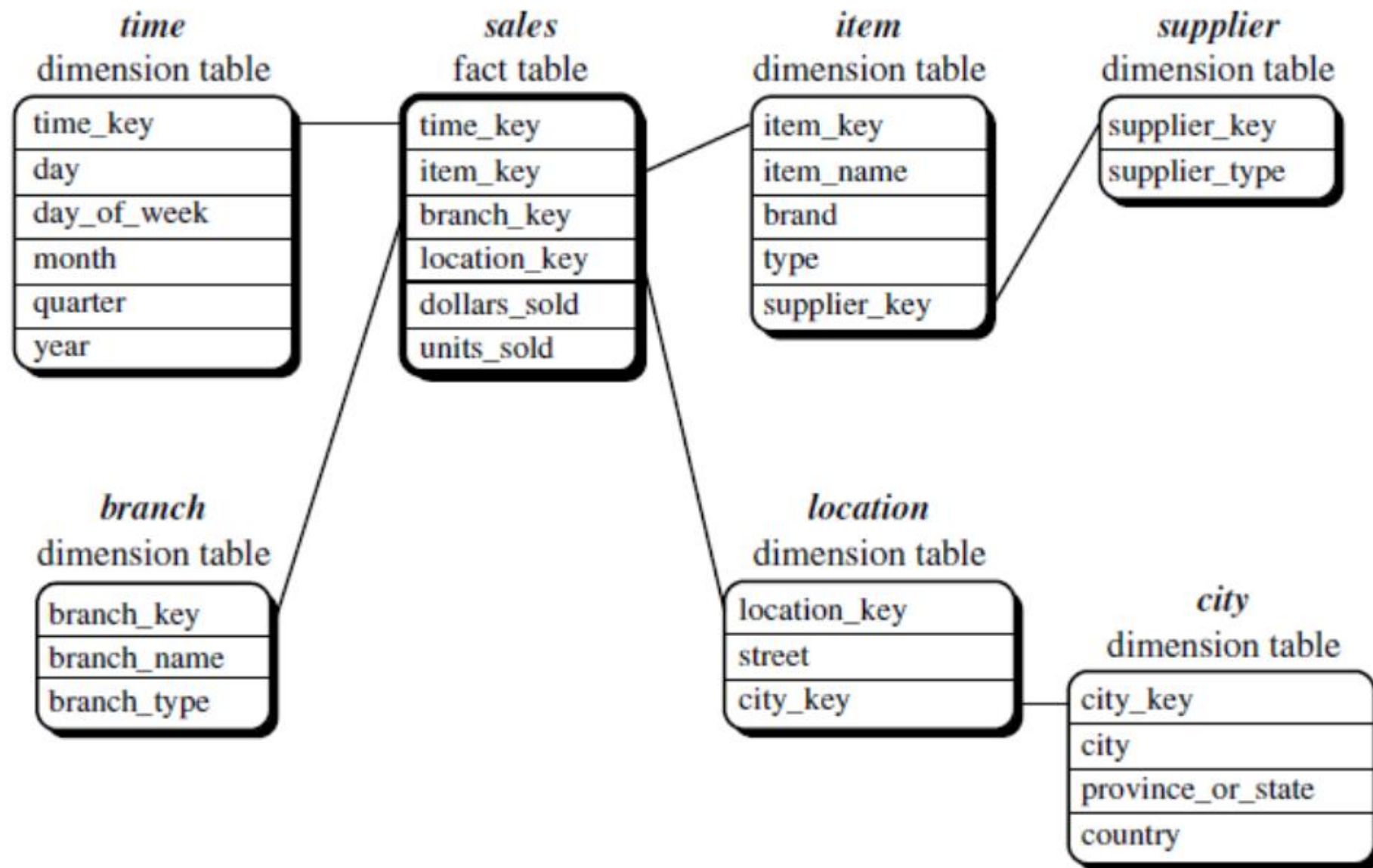
- A **star schema** is used to model **AllElectronics sales data**.
- Sales are analyzed across **four dimensions**:
 - **Time**
 - **Item**
 - **Branch**
 - **Location**
- The schema has a **central fact table (Sales)**.
- The **fact table** contains:
 - **Foreign keys** to each of the four dimension tables
 - **Measures** such as **dollars sold** and **units sold**
- To **reduce the size of the fact table**, **system-generated dimension identifiers** are used (e.g., time key, item key).
- Each **dimension is represented by a single table** in a star schema.
- Each **dimension table contains multiple attributes** related to that dimension.

- **Example:** The **Location dimension table** includes attributes such as:
 - location key
 - street
 - city
 - state
 - country
- This design may cause **attribute redundancy** within dimension tables.
- **Example of redundancy:**
 - Cities like **Vancouver** and **Victoria** share the same **province (British Columbia)** and **country (Canada)**.
- **Attributes in a dimension table can form:**
 - A **hierarchy (total order)**, or
 - A **lattice (partial order)** for flexible analysis.



Snowflake Schema

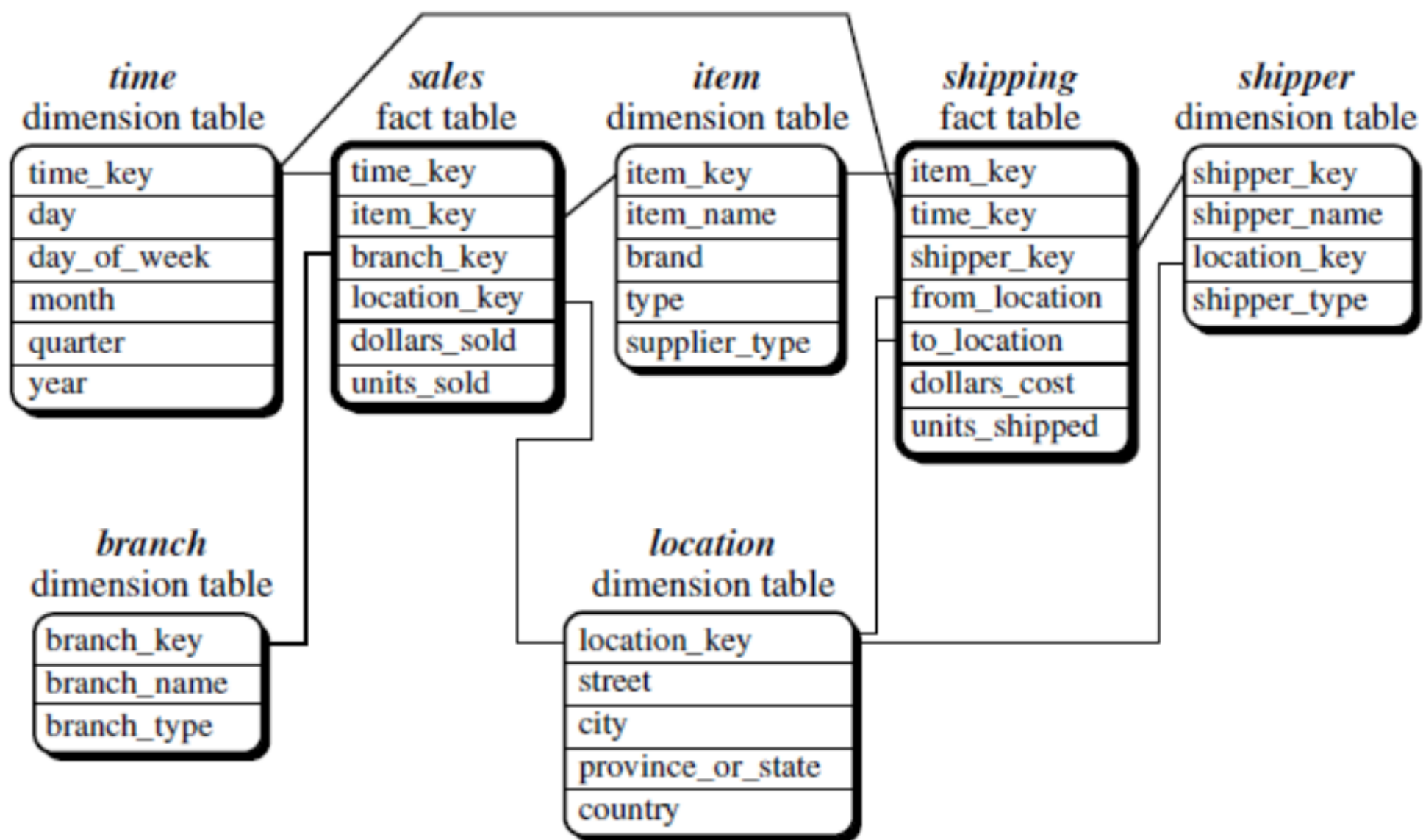
- A **snowflake schema** is a refined form of the **star schema**.
- The **sales fact table** is **identical to the star schema**.
- The main difference is in the **design of dimension tables**.
- In a snowflake schema, **dimension tables are normalized** to reduce redundancy.
- The **item dimension** from the star schema is split into:
 - **Item table**: item key, item name, brand, type, supplier key
 - **Supplier table**: supplier key, supplier type
- The **supplier key** in the item table links to the **supplier dimension table**.
- The **location dimension** is normalized into:
 - **Location table**
 - **City table**
- The **city key** in the location table links to the **city dimension table**.
- Further normalization can be applied to **province/state** and **country** tables.
- Snowflake schema **minimizes data redundancy**.
- It results in a **more complex structure** and **additional joins** compared to a star schema.



3. Fact Constellation Schema

- A **fact constellation schema** contains **multiple fact tables** instead of a single fact table.
- In the given example, the schema includes **two fact tables**:
 - Sales
 - Shipping
- The **sales fact table** is **identical to the sales table in the star schema**.
- The **shipping fact table** includes:
 - **Five dimensions (keys)**:
 - item key
 - time key
 - shipper key
 - from location
 - to location
 - **Two measures**:
 - dollars cost
 - units shipped

- A key feature of fact constellation schema is that **dimension tables can be shared** among multiple fact tables.
- In this example, the **time, item, and location dimension tables** are shared by both **sales** and **shipping** fact tables.
- **Data Warehouse vs Data Mart:**
 - A **data warehouse** stores enterprise-wide data covering multiple subjects such as customers, items, sales, assets, and personnel.
 - **Fact constellation schema** is commonly used in data warehouses because it supports **multiple interrelated subjects**.
 - A **data mart** is a **department-level subset** of a data warehouse focusing on selected subjects.
 - **Star schema** and **snowflake schema** are commonly used for data marts since they model **single subjects**.
 - Among them, the **star schema** is more **popular and efficient**.



- **Measures: Their Categorization and Computation**
- Measures answer the question: “**How are measures computed?**”
- A **multidimensional data cube point** is defined by a set of **dimension–value pairs**
 - Example: $\langle \text{time} = \text{Q1}, \text{location} = \text{Vancouver}, \text{item} = \text{computer} \rangle$
- A **data cube measure** is a **numerical function** evaluated at each point in the data cube.
- Measure values are computed by **aggregating data** that match the given dimension–value pairs.
- Measures are classified into **three categories** based on the type of aggregate function used:
 - **Distributive**
 - **Algebraic**
 - **Holistic**

- **Distributive Measures**
- An aggregate function is **distributive** if it can be computed in a **distributed manner**.
- The dataset can be **partitioned into multiple subsets**.
- The aggregate function is applied to each subset separately.
- The **final result**, obtained by combining subset results, is the **same as applying the function to the entire dataset**.
- Example:
 - **count()**:
 - Compute count for each subcube.
 - Sum all counts to get the final result.
- Hence, **count()** is a distributive aggregate function.
- Other examples of distributive aggregate functions:
 - **sum()**
 - **min()**
 - **max()**
- A **measure is distributive** if it is computed using a distributive aggregate function.
- Distributive measures are **efficient to compute** and **well-suited for large data cubes**.

Extraction, Transformation and Loading (ETL)

- **ETL** is a core process in a **data warehouse** that involves collecting data from different sources, converting it into a suitable format, and storing it in the data warehouse for analysis.

1. Extraction

- **Extraction** is the process of **collecting data from various source systems**.
- **Key Points**
- Data can be extracted from **databases, flat files, spreadsheets, APIs, logs, or external systems**.
- Source systems are usually **OLTP systems**.
- Extraction can be:
 - **Full extraction** – entire data is extracted each time.
 - **Incremental extraction** – only new or changed data is extracted.
- **Example**
- Extract customer details from:
 - MySQL database (Customer table)
 - Excel file containing sales data
 - CRM system storing customer interactions

2. Transformation

Transformation is the process of **cleaning, converting, and integrating data** into a consistent format suitable for the data warehouse.

Key Points

- Removes **errors, duplicates, and inconsistencies**.
- Converts data types and formats.
- Applies **business rules**.
- Combines data from multiple sources.
- Generates **derived or calculated fields**.

Common Transformation Tasks

- Data cleaning (removing null or invalid values)
- Data normalization
- Aggregation and summarization
- Encoding and decoding values
- Surrogate key generation

Example

- Convert date format from `DD-MM-YYYY` to `YYYY-MM-DD`.
- Standardize gender values:
 - "M", "Male" → "Male"
 - "F", "Female" → "Female"
- Calculate **total sales** = **quantity** × **price**.
- Remove duplicate customer records.

3. Loading

Loading is the process of **storing the transformed data into the data warehouse.**

Key Points

- Data is loaded into **fact tables and dimension tables.**
- Loading can be:
 - **Full load** – entire dataset is loaded.
 - **Incremental load** – only new or updated records are loaded.
- Loading strategies include:
 - Append
 - Update
 - Replace

Example

- Load cleaned customer data into the **Customer Dimension Table.**
- Load aggregated sales data into the **Sales Fact Table.**
- Update existing records with new values during incremental loads.

ETL Process Flow

- Extract daily sales data from:
 - Point of Sale systems
 - Online shopping database
- Transform the data by:
 - Removing duplicate entries
 - Converting currencies to INR
 - Calculating total sales and profit
- Load the final data into:
 - Sales fact table
 - Time, Product, and Store dimension tables

Importance of ETL

- Ensures **data quality and consistency**.
- Integrates data from **multiple heterogeneous sources**.
- Supports **accurate reporting and decision-making**.
- Improves performance of **OLAP queries**.

Meta Data Repository

- **Metadata** means *data about data*. In a data warehouse, metadata defines and describes warehouse objects.
- Metadata includes **data names, definitions, source details, time stamps**, and information added during **data cleaning and integration**.

Contents of a Metadata Repository

1. Structural Metadata

- Describes the structure of the data warehouse.
- Includes schema, views, dimensions, hierarchies, derived data definitions, and data mart locations and contents.

2. Operational Metadata

- Contains data lineage (history and transformations of data).
- Includes data status (active, archived, purged) and monitoring information such as usage statistics, error reports, and audit trails.

3. Summarization and Aggregation Metadata

- Stores algorithms for measures and dimensions.
- Includes details on granularity, partitions, subject areas, aggregation, summarization, and predefined queries and reports.

4. Mapping Metadata

- Defines mapping from operational systems to the data warehouse.
- Includes source databases, extraction methods, transformation and cleaning rules, refresh and purging rules, and security details.

5. Performance Metadata

- Contains information on indexes and profiles for efficient data access.
- Includes scheduling rules for data refresh, update, and replication cycles.

6. Business Metadata

- Stores business terms and definitions.
- Includes data ownership details and charging or usage policies.

Uses of a Metadata Repository

- **Centralized Data Management:** Consolidating all dataset information into one system streamlines the management and retrieval of data.
- **Tracking Data History:** Metadata repositories track when and how datasets are **modified**, providing a **historical** record of changes.
- **Facilitating Compliance and Governance:** They help organizations maintain control over data usage, ensuring **compliance** with regulations and governance policies.
- **Enhancing User Accessibility to Datasets:** Metadata repositories improve data **accessibility**, allowing users to quickly find and access the data they need.

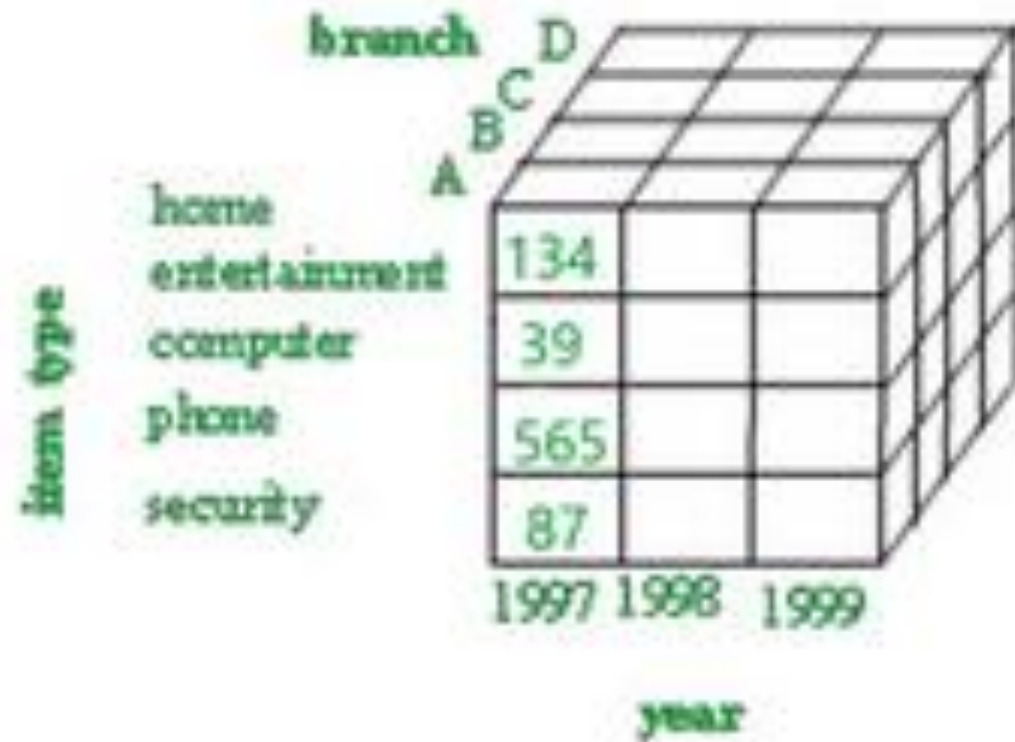
Features of a Metadata Repository

- **Data Storage and Organization:** A well-structured repository makes it easy to **categorize** and manage datasets.
- **Tracking of Data Origin and Modifications:** Each dataset's journey is **logged**, helping users trace its origin and any modifications made.
- **User Access and Permission Management:** Repositories **control** who can access or edit data, ensuring **security** and proper data handling.
- **Historical Data Tracking:** Detailed records of dataset **updates** and **changes** allow users to go back and understand how data has evolved.

Data Cube or OLAP approach in Data Mining

What is OLAP?

- **OLAP** stands for **Online Analytical Processing**.
- It is a technology used for **multidimensional analysis of business data**.
- OLAP provides **interactive and fast access** to large volumes of data.
- It supports **complex calculations, aggregations, and analytical queries**.
- OLAP is widely used in **Business Intelligence (BI)** and **decision-making systems**.
- In data warehousing, data is organized in the form of **data cubes**.
- A **data cube** represents data in a **multidimensional matrix** format.
- Each dimension contains **multiple attributes**, such as time, product, or location.
- The data cube represents a **high-dimensional space** for analysis.
- It visually shows how different data attributes are **arranged and related** in the data model.
- Data cubes help users analyze data from **multiple perspectives efficiently**.

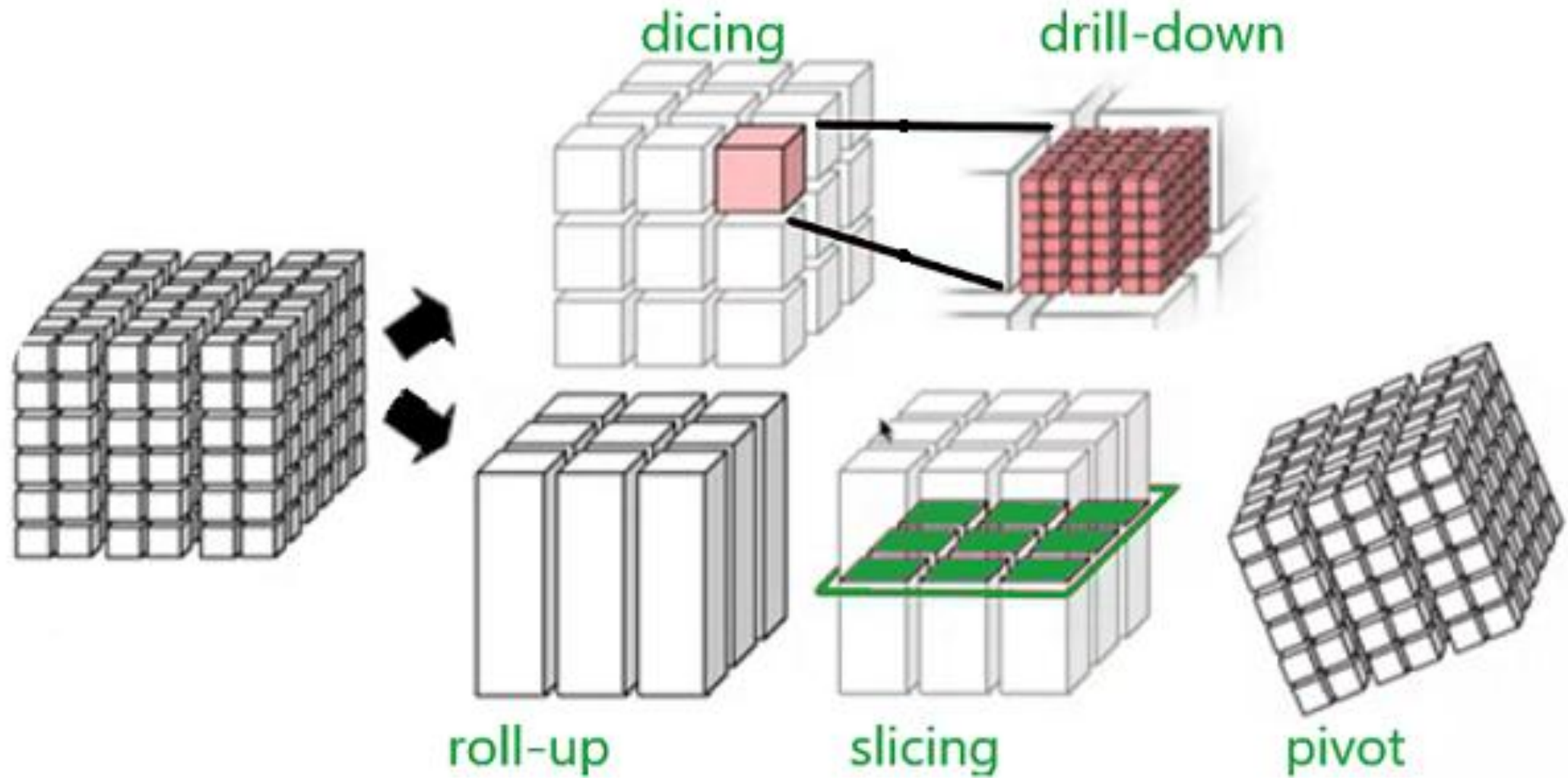


The example above is a 3D cube having attributes like branch(A,B,C,D), item type(home,entertainment,computer,phone,security), year(1997,1998,1999) .

Data cube classification

- *The data cube can be classified into two categories:*
- **Multidimensional data cube:** It basically helps in storing large amounts of data by making use of a multi-dimensional array. It increases its efficiency by keeping an index of each dimension. Thus, dimensional is able to retrieve data fast.
- **Relational data cube:** It basically helps in storing large amounts of data by making use of relational tables. Each relational table displays the dimensions of the data cube. It is slower compared to a Multidimensional Data Cube.

Data cube operations



(a) Roll-Up

- Summarizes data by **moving up the dimension hierarchy**.
- Example:
 - City-level sales → State-level sales → Country-level sales.

(b) Drill-Down

- Opposite of roll-up; moves from **summary to detailed data**.
- Example:
 - Yearly sales → Quarterly sales → Monthly sales.

(c) Slice

- Selects a **single dimension value**, creating a sub-cube.
- Example:
 - Sales data for the year **2025** only.

(d) Dice

- Selects data using **multiple dimension values**.
- Example:
 - Sales of *Laptops* and *Mobiles* in *Tamil Nadu* during *Q1* and *Q2*.

(e) Pivot (Rotate)

- Reorients the data cube to view data from **different perspectives**.
- Example:
 - Viewing sales by product vs by time.

Advantages of data cubes

- **Multi-dimensional analysis:** Data cubes enable multi-dimensional analysis of business data, allowing users to view data from different perspectives and levels of detail.
- **Interactivity:** Data cubes provide interactive access to large amounts of data, allowing users to easily navigate and manipulate the data to support their analysis.
- **Speed and efficiency:** Data cubes are optimized for OLAP analysis, enabling fast and efficient querying and aggregation of data.
- **Data aggregation:** Data cubes support complex calculations and data aggregation, enabling users to quickly and easily summarize large amounts of data.
- **Improved decision-making:** Data cubes provide a clear and comprehensive view of business data, enabling improved decision-making and business intelligence.
- **Accessibility:** Data cubes can be accessed from a variety of devices and platforms, making it easy for users to access and analyze business data from anywhere.
- Helps in giving a summarised view of data.
- Data cubes store large data in a simple way.
- Data cube operation provides quick and better analysis,
- Improve performance of data.

Disadvantages of data cube

- **Complexity:** OLAP systems can be complex to set up and maintain, requiring specialized technical expertise.
- **Data size limitations:** OLAP systems can struggle with very large data sets and may require extensive data aggregation or summarization.
- **Performance issues:** OLAP systems can be slow when dealing with large amounts of data, especially when running complex queries or calculations.
- **Data integrity:** Inconsistent data definitions and data quality issues can affect the accuracy of OLAP analysis.
- **Cost:** OLAP technology can be expensive, especially for enterprise-level solutions, due to the need for specialized hardware and software.
- **Inflexibility:** OLAP systems may not easily accommodate changing business needs and may require significant effort to modify or extend.

Data Warehouse Design and Usage

- Designing a data warehouse involves selecting an appropriate **design approach** that defines how the system is **structured, developed, and scaled**.
- The chosen approach directly affects **data consistency, system performance, integration complexity, and speed of insight delivery**.
- A well-chosen design approach ensures **efficient data management and reliable analytics** across teams.
- There are **two common approaches** used for constructing a data warehouse.

- **Top-Down Approach:** This method starts with designing the overall data warehouse architecture first and then creating individual data marts.
- **Bottom-Up Approach:** In this method, data marts are built first to meet specific business needs and later integrated into a central data warehouse.

Components of Data Warehouse Architecture

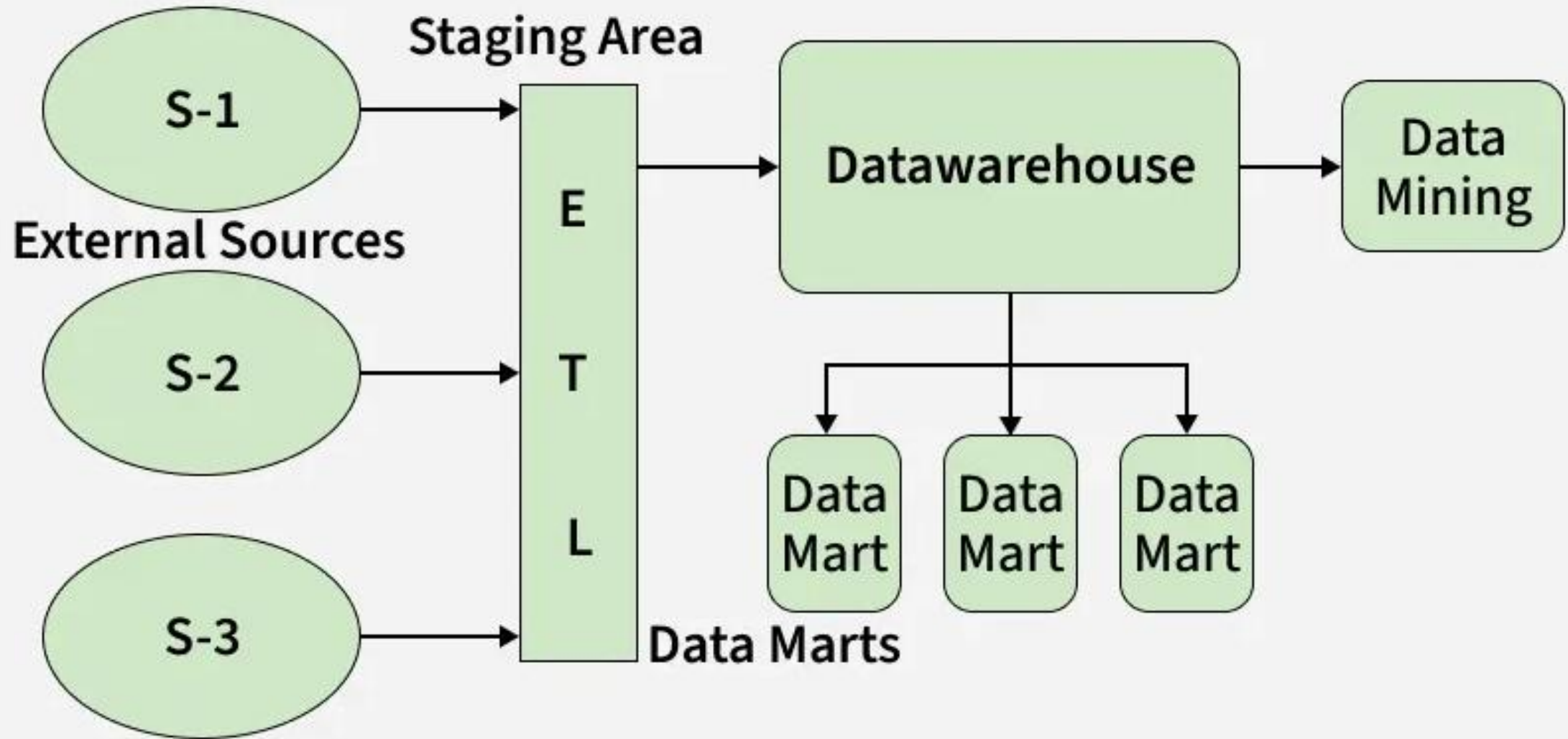
- A data warehouse architecture consists of several key components that work together to store, manage and analyze data.
- **External Sources:** Where data originates. Includes: Structured (databases, spreadsheets), Semi-structured (XML, JSON) & Unstructured (emails, images)
- **Staging Area:** A temporary space where raw data is cleaned and validated before moving to the warehouse. ETL tools manage this process: Extract (E) - Pulls raw data from sources, Transform (T) - Standardizes and formats the data & Load (L) - Moves the data into the data warehouse
- **Data Warehouse:** A central storage for organized, cleansed data, including both raw data and metadata. Supports analysis, reporting and decision-making.
- **Data Marts:** Smaller, focused sections of the data warehouse for specific teams (e.g., sales, marketing), enabling quick access to relevant data.
- **Data Mining:** Analyzing large datasets in the warehouse to find patterns, trends and insights that support decisions and improve operations.

1. Top-Down Approach

- The Top-Down Approach, introduced by Bill Inmon, is a method for designing data warehouses that starts by building a centralized, company-wide data warehouse.
- This central repository acts as the single source of truth for managing and analyzing data across the organization. It ensures data consistency and provides a strong foundation for decision-making.

Working of Top-Down Approach

- **Central Data Warehouse:** The process begins with creating a comprehensive data warehouse where data from various sources is collected, integrated and stored. This involves the ETL (Extract, Transform, Load) process to clean and transform the data.
- **Specialized Data Marts:** Once the central warehouse is established, smaller, department-specific data marts (e.g., for finance or marketing) are built. These data marts pull information from the main data warehouse, ensuring consistency across departments.



Advantages of Top-Down Approach

- **Consistent View:** Data marts built from a central warehouse ensure uniform data across departments, reducing reporting discrepancies.
- **High Data Consistency:** Standardizing data through one source minimizes errors and improves the reliability of insights.
- **Simplified Maintenance:** Updates in the central warehouse automatically reflect in all data marts, saving time and effort.
- **Scalable Architecture:** New data marts can be added easily as business needs grow or change.
- **Stronger Governance:** Centralized control improves data security, access management and compliance.
- **Less Data Duplication:** Data is stored once in the warehouse, saving space and avoiding redundant or conflicting records.
- **Better Reporting:** A unified data source enables faster, more accurate reporting and decision-making.
- **Improved Integration:** Central sourcing makes it easier to combine data from multiple systems for deeper analysis.

Disadvantages of Top-Down Approach

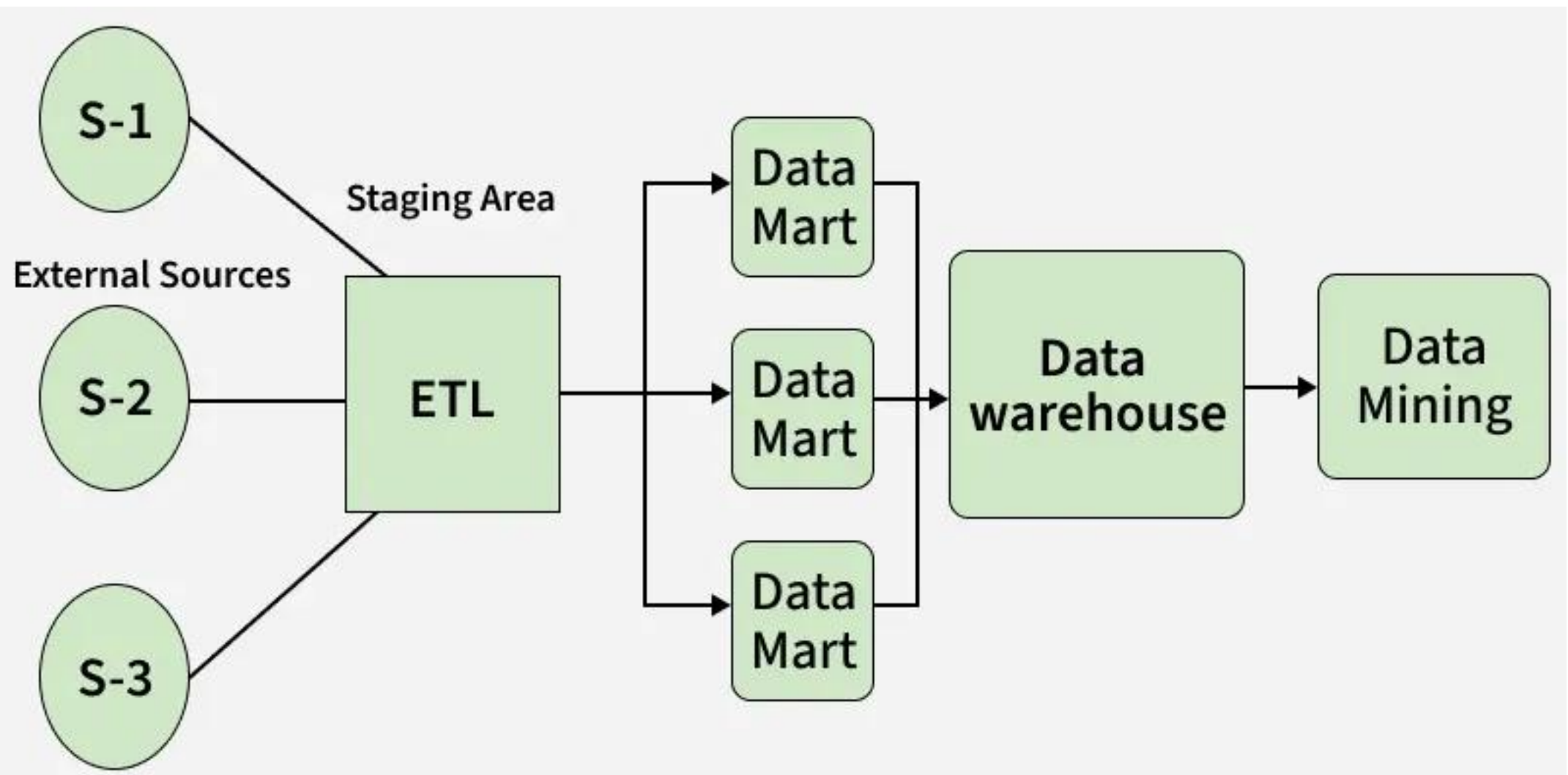
- **High Cost & Time:** Building a central data warehouse and data marts requires major investment and long implementation time, making it hard for smaller organizations.
- **Complex Setup:** Designing and managing a centralized system is technically complex and requires skilled resources and careful planning.
- **Low Flexibility:** Predefined structures make it hard to adapt quickly to changing business needs or reporting requirements.
- **Limited User Input:** IT-led development can exclude business users, resulting in solutions that may not meet their actual needs.
- **Data Delays:** Pulling data from various systems can cause processing delays, affecting real-time reporting and insights.
- **Unclear Data Ownership:** Centralization can blur responsibility, making it unclear who manages or maintains specific data.

2. Bottom-Up Approach

- The Bottom-Up Approach, popularized by Ralph Kimball, takes a more flexible and incremental path to designing data warehouses.
- Instead of starting with a central data warehouse, it begins by building small, department-specific data marts that cater to the immediate needs of individual teams, such as sales or finance.
- These data marts are later integrated to form a larger, unified data warehouse.

Working of Bottom-Up Approach

- **Department-Specific Data Marts:** The process starts with creating data marts for individual departments or specific business functions. These data marts are designed to meet immediate data analysis and reporting needs, allowing departments to gain quick insights.
- **Integration into a Data Warehouse:** Over time, these data marts are connected and consolidated to create a unified data warehouse. The integration ensures consistency and provides a comprehensive view of the organization's data.



Advantages of Bottom-Up Approach

- **Faster Reporting:** Data marts allow quick insights and report generation.
- **Step-by-Step Development:** Enables gradual rollout with quick wins.
- **User-Centric:** Involves business users to meet actual needs.
- **Highly Flexible:** Easily customized for departments or evolving needs.
- **Quick Results:** Early setup gives immediate value.

Disadvantages of Bottom-Up Approach

- **Inconsistent Views:** Different structures can lead to inconsistent reporting.
- **Data Silos:** Independent marts may cause duplication and isolation.
- **Integration Difficulty:** Combining varied marts into one warehouse is hard.
- **Redundant Efforts:** Similar marts may be built by different teams.
- **Harder to Manage:** Multiple marts increase maintenance overhead.
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